

CTOC · COMMAND

OPERATIONS

Fusion Kill Chain

7

Threat Detection

218

Incidents & Cases

14

Analyst Workload

8

Automation

3.8K

DOMAINS

Data Loss Prev.

412

Threat Intel

42K

Money Movement

142

Fraud Monitoring

312

Vulnerabilities

38

Insider Threat

22

Executive Travel

22

REPORTING

Executive Overview

SHIFT · A · DAY

M. Okafor · T2

12 open · 92% SLA

VIEW / OPS / FUSION KILL CHAIN

RANGE

LIVE

1H

24H

7D

30D

SEARCH

FILTERS

SAVED VIEWS

UTC 22:28:57

Fusion Kill Chain

ACTIVE CAMPAIGNS · CROSS-STREAM CORRELATION · MITRE ATT&CK ALIGNMENT 3 CRITICAL · 4 HIGH

ACTIVE CAMPAIGNS

7 ▲ +2

2 escalated past 06:00 UTC

OPEN ALERTS

218 ▼ -14

1,442 ingested last hr

CRITICAL INCIDENTS

3 ▲ +1

INC-9981 ETA contain 14:55

ASSETS COMPROMISED

12 ▲ +3

7 production · 3 corp · 2 cloud

MEAN TIME TO DETECT

4m ▼ -18s

Target ≤ 5m

MEAN TIME TO RESPOND

11m ▼ -02m

Target ≤ 15m

AUTO-CONTAINED

83% ▲ +4%

42 of 51 last hour

LOCKHEED CYBER KILL CHAIN STAGE 5 SELECTED · INSTALLATION

WINDOW 1H 24H 7D 30D

STAGE 01
Reconnaissance

EVENTS 142

ACTIVE 0

CONF. 42%

STAGE 02
Weaponization

EVENTS 23

ACTIVE 0

CONF. 61%

STAGE 03
Delivery

EVENTS 87

ACTIVE 1

CONF. 74%

STAGE 04
Exploitation

EVENTS 14

ACTIVE 2

CONF. 88%

STAGE 05
Installation

EVENTS 6

ACTIVE 2

CONF. 91%

STAGE 06
C2

EVENTS 3

ACTIVE 1

CONF. 93%

STAGE 07
Actions on Objectives

EVENTS 2

ACTIVE 1

CONF. 96%

ACTIVE CAMPAIGNS CORRELATED · LAST 72H

7 ROWS FILTER

ID

CAMPAIGN

ACTOR

STAGE

SEV

CONF

ASSETS

MTTD

CMP-2841

Operation TIDEWAVE
Credential harvesting → DC lateral → S3 staging

APT-441 (Nostromo)

6/7

CRITICAL

93%

7

00:04:12

CMP-2839

Operation RUSTFALL
Office macro chain targeting Treasury workstations

FIN-209 (Khepri)

5/7

HIGH

81%

3

00:11:55

CMP-2836

Operation PALEHORSE
Anomalous RDP from Tier-O admin account

Unattributed

4/7

HIGH

68%

2

00:24:08

CMP-2832

Operation IRONBLOOM
Phishing wave targeting Wholesale Banking ops

eCrime cluster

3/7

MEDIUM

55%

12

01:02:44

CMP-2829

Operation HEARTSTONE
External recon: subdomain enumeration + cert mining

APT-209 (Vanta)

2/7

MEDIUM

47%

0

02:14:01

CMP-2825

Operation CLEARWATER
Privileged user —

Insider (suspected)

5/7

CRITICAL

74%

1

04:08:20

BLAST RADIUS · TOPOLOGY

CMP-2841 · OPERATION TIDEWAVE

RESET OPEN

TOPOLOGY · 9 ASSETS · 8 FLOWS · CMP-2841

VPN-GW

PROD-DB-01

PROD-DB-02

IDP-RING

APP-77

WS-441

EDGE-LB-A

DC-01

OBJ-STORE

EXFIL PATH

ESCALATION

LATERAL

ENTRY VECTOR

TACTIC

STARTED

CONFIDENCE

Phishing → macro · WS-441

TA0011 · Exfiltration

2026-05-25T18:42Z

93% · Bayesian fusion v3

01 · THE CHALLENGE

Modern security operations don't fail from lack of data, they fail from *too much of it*.

An enterprise bank's Cyber Threat Operations Center monitors petabytes of telemetry across dozens of disconnected tools. Analysts triage 1,400+ alerts an hour. Leadership wants one number that tells them whether the bank is safe today.

The pattern this concept responds to is familiar: tier-one analysts swivel-chairing between six browser tabs to correlate a single phishing event. Critical alerts buried by noisy ones. Inconsistent severity definitions between detection tools. Executives reading PDF reports that were stale the moment they were exported.

The fragmentation wasn't a tooling problem, every individual product worked. It was an **interaction problem**: thirteen dashboards built by thirteen vendors, each speaking a different visual dialect, and a workforce paying the cognitive tax.

●●●●●●●● → 6 TABS TO CORRELATE ONE EVENT → 1 SYSTEM

PROBLEM SIGNALS

TOOL SPRAWL

Analysts lose minutes per shift switching tools to confirm what one tool already surfaced.

INVISIBLE WORKLOAD

Managers have no live view of team capacity, so burnout is discovered late.

SILOLED QUEUES

Cyber and fraud queues don't share context, the same actor gets investigated twice.

EXEC TRANSLATION GAP

Leadership wants one answer, are we safe right now? not another graph to interpret.

"A SOC analyst's most expensive resource isn't time, it's the *working memory* they spend reconstructing context that the tools already had."

THE DESIGN THESIS BEHIND THE CONCEPT

02 · FRAMING & DOMAIN RESEARCH

Grounding the concept in how security operations *actually work*.

This is a self-directed concept, so the research was desk-based: I studied the source PRD, published SOC operating models, MITRE and Lockheed Martin frameworks, and the way existing security tools present information. The goal was to design *as if* for four real constituencies: **tier-1, tier-2, and tier-3 analysts, SOC and CTOC managers, fraud and incident response teams, and cybersecurity leadership** who report to the board.

Three questions framed the work: what are the viewing patterns (how would a tier-1 analyst look at a wallboard?), what are the workflow behaviors (the sequence between detection and containment?), and what are the environmental constraints (dark control rooms, 24/7 shift rotation, multi-monitor setups)?

WORKING PREMISES

GLANCEABILITY	A wallboard has ~two seconds to communicate whether it matters. If it doesn't convey severity instantly, it gets tuned out.
ALERT FATIGUE	High alert volume is a known driver of analyst burnout, treated here as a design problem, not just an HR one.
CONTEXT DECAY	Investigation context is lost every time an analyst switches tools. The product should carry context with the analyst.
EXEC TRANSLATION	Leadership doesn't need simpler dashboards, it needs dashboards that tell a coherent story in plain English.

SOC · SECURITY OPERATIONS CENTER

The team and the room responsible for monitoring, detecting, and responding to cybersecurity threats in real time. Typically staffed 24/7 in tiered shifts.

CTOC · CYBER THREAT OPERATIONS CENTER

A SOC at enterprise scale, combining cyber threat detection with fraud monitoring, insider threat, threat intelligence, and executive risk reporting into a single function.

TIER 1 / 2 / 3 · ANALYST SENIORITY

T1 triages alerts. **T2** investigates and escalates. **T3** leads incident response and threat hunting. Each tier has different information needs.

WALLBOARD

A large-format dashboard mounted on a wall in the SOC, visible to the whole team. Optimized for glanceability at distance, not interaction.

MTTD · MTTR

Mean Time to Detect / Respond. The two headline operational metrics for a SOC. Lower is better. Both anchor the concept's design goals.

03 · PERSONAS WE DESIGNED FOR

Three jobs, three latencies, *three layouts*.

The product had to feel native to a tier-1 analyst working an eight-hour overnight shift and to a CISO presenting to the board fifteen minutes later. I resolved that tension by designing **three distinct experience layers** against three modeled persona types.

PERSONA 01

SOC Analyst

PERSONA 02

SOC Manager · CTOC Lead

PERSONA 03

Executive Leadership

Triages alerts and investigates incidents under time pressure. Lives in the dense analyst dashboard.

PAIN POINTS

Alert fatigue · noise · tool swivel-chairing · dense unreadable tables · lost investigation context

NEEDS

Fast triage workflows · glanceable severity · real-time updates · persistent investigation state

REFRESH TOLERANCE

Real-time / sub-second

Oversees the floor. Balances workload, tracks SLA performance, escalates major incidents.

PAIN POINTS

No view of team capacity · burnout detection is reactive · SLA breaches found after the fact

NEEDS

Team workload visibility · queue health at a glance · escalation pathways · shift-level performance data

REFRESH TOLERANCE

Near real-time (≤ 30 s)

CISO, CRO, board reporting. Translates technical posture into business risk and regulatory exposure.

PAIN POINTS

Stale exported reports · technical jargon · disconnected risk frameworks

NEEDS

Plain-English summaries · business impact framing · trend visualization · audit-ready evidence trail

REFRESH TOLERANCE

5-15 min · per-meeting

04 · DESIGN PRINCIPLES

Seven principles. Every screen had to *defend itself* against all of them.

Before we drew a single screen, we wrote down what the product should never do. These principles became the criteria every design review used. When a panel didn't earn its place against them, it was cut.

01

Glanceability first

If a wallboard takes more than two seconds to read, it has failed. Critical state must dominate visual attention without the viewer choosing to look for it.

02

Severity is obvious

A five-level severity ladder, color- and shape-encoded, applied consistently across every screen. No exceptions, no per-tool dialects.

03

Reduce cognitive load

Progressive disclosure: the dashboard shows what's true, the drawer shows why, the timeline shows when, the playbook shows how to act.

04

Optimize for stress

The interface during a P1 incident should feel calmer, not louder. Layout density stays constant; color and motion carry the urgency.

05

Consistency creates speed

Shared interaction patterns, same filter chips, same drawer, same severity pill, so an analyst trained on one dashboard already knows the next.

06

Motion clarifies

Animation is used only to signal change: a new critical alert pulses; a contained event fades. Decoration is forbidden.

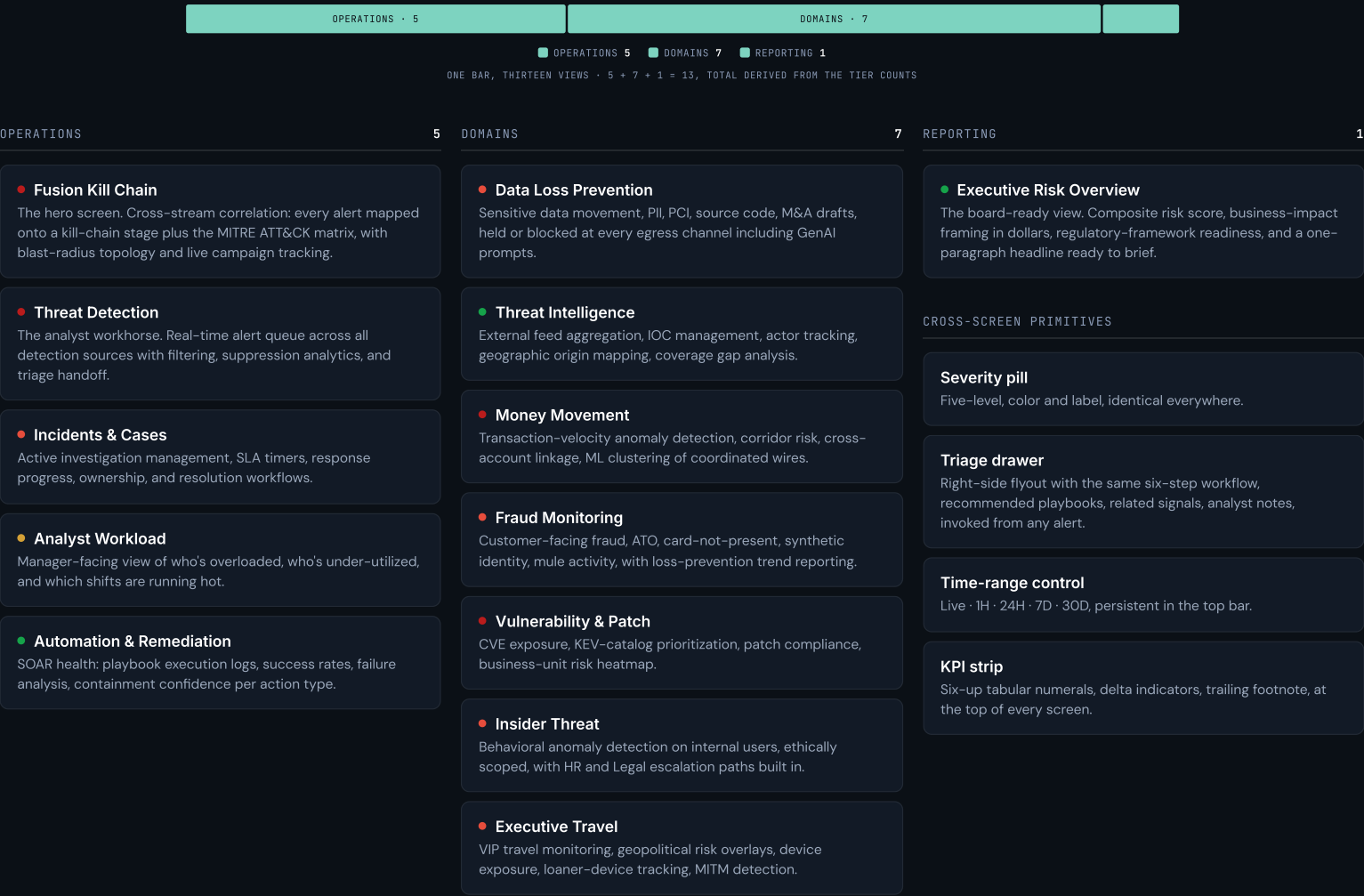
07

Density without chaos

The product is information-dense by design, but information density is earned through hierarchy, not crowded by stacking.

Thirteen dashboards, three operational tiers, *one design language.*

The product is organized into three tiers that mirror how the work actually flows: **Operations** (live triage and response), **Domains** (specialized monitoring), and **Reporting** (executive translation). Every dashboard inherits the same chrome, severity model, and interaction grammar, what changes is information density and refresh cadence.



What's the worst thing *happening to us right now?*

The Fusion Kill Chain is the dashboard the team turns on first thing in the morning and leaves on the wallboard all day. Its job is to fuse signals from every detection source, endpoint, network, identity, cloud, email, fraud, onto a single attack-stage timeline, then make the highest-priority correlation visible inside two seconds.

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VIEW / OPS / FUSION KILL CHAIN

RANGE

LIVE

1H

24H

7D

30D

SEARCH

FILTERS

SAVED VIEWS

THEME

LIGHT

DARK

UTC 14:42:08

Fusion Kill Chain

ACTIVE CAMPAIGNS · CROSS-STREAM CORRELATION · MITRE ATT&CK ALIGNMENT 3 CRITICAL · 4 HIGH

CORRELATION RULES

EXPORT IOCS

DECLARE INCIDENT

ACTIVE CAMPAIGNS 7 ▲ +2 2 escalated past 06:00 UTC	OPEN ALERTS 218 ▼ -14 1,442 ingested last hr	CRITICAL INCIDENTS 3 ▲ +1 INC-9981 ETA contain 14:55	ASSETS COMPROMISED 12 ▲ +3 7 production · 3 corp · 2 cloud	MEAN TIME TO DETECT 4m ▼ -18s Target ≤ 5m	MEAN TIME TO RESPOND 11m ▼ -02m Target ≤ 15m	AUTO-CONTAINED 83% ▲ +4% 42 of 51 Last hour
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LOCKHEED CYBER KILL CHAIN

STAGE 5 SELECTED · INSTALLATION

WINDOW

1H

24H

7D

30D

STAGE 01 Reconnaissance EVENTS 142 ACTIVE 0 CONF. 42%	STAGE 02 Weaponization EVENTS 23 ACTIVE 0 CONF. 61%	STAGE 03 Delivery EVENTS 87 ACTIVE 1 CONF. 74%	STAGE 04 Exploitation EVENTS 14 ACTIVE 2 CONF. 88%	STAGE 05 Installation EVENTS 6 ACTIVE 2 CONF. 91%	STAGE 06 C2 EVENTS 3 ACTIVE 1 CONF. 93%	STAGE 07 Actions on Objectives EVENTS 2 ACTIVE 1 CONF. 96%
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ACTIVE CAMPAIGNS

CORRELATED · LAST 72H

7 ROWS

FILTER

ID	CAMPAIGN	ACTOR	STAGE	SEV	CONF	ASSETS	MTTD
CMP-2841	Operation TIDEWAVE Credential harvesting → DC lateral → S3 staging	APT-441 (Nostramo)	6/7	CRITICAL	93%	7	00:04:12
CMP-2836	Operation PALEHORSE Anomalous RDP from Tier-0 admin account	Unattributed	4/7	HIGH	68%	2	00:24:08
CMP-2832	Operation IRONBLOOM Phishing wave targeting Wholesale Banking ops	eCrime cluster	3/7	MEDIUM	55%	12	01:02:44
CMP-2829	Operation HEARTSTONE External recon: subdomain enumeration + cert mining	APT-209 (Vanta)	2/7	MEDIUM	47%	0	02:14:01
CMP-2825	Operation CLEARWATER Privileged user — atypical data access patterns	Insider (suspected)	5/7	CRITICAL	74%	1	04:08:20
CMP-2822	Operation SILVERMOTH Failed credential stuffing — IPs blocked at edge	FIN-318	3/7	LOW	33%	0	—

MITRE ATT&CK · ENTERPRISE

CMP-2841 MAPPED · 18 TECHNIQUES OBSERVED

HEATMAP

TIMELINE

EXPORT NAVIGATOR

INITIAL ACCESS	EXECUTION	PERSISTENCE	PRIV. ESCALATION	DEFENSE EVASION	LATERAL MOVMT.	EXFILTRATION
T1566 Phishi... 14 T1190 Public-Facing A... 6 T1078 Valid Accou... 3 T1133 External Rem... T1200 Hardware A...	T1059 Cmd & Scr... 22 T1204 User Execut... 8 T1053 Scheduled T... 2 T1129 Shared Modu... T1585 Server Compon...	T1547 Boot/Logon A... 4 T1098 Account Man... 2 T1136 Create Accou... 1 T1595 Server Compon...	T1548 Abuse Elevati... 3 T1068 Exploit Priv. ... 1 T1055 Process Injecti... T1140 Deobfusc...	T1027 Obfuscated Fil... 11 T1070 Indicator Remo... 4 T1036 Masqueradi... 2 T1534 Internal Sp...	T1021 Remote Servi... 7 T1350 Alt. Au... 2 T1534 Internal Sp...	T1041 C2 Channel E... 2 T1567 Web Service E... 1 T1029 Scheduled X...

INVESTIGATION TIMELINE

CROSS-STREAM EVENTS · LAST 24M

FILTER

EXPAND

14:42:08 UTC

LSASS credential access on APP-77

EDR rule R-CB-0144 · MITRE T1003 · auto-contained at 14:42:24

14:41:52 UTC

DCSync from non-DC origin (DC-01)

NDR/Vectra · attributed to CMP-2841 · K. Brennan paged

14:39:18 UTC

2.3GB anomalous egress · OBJ-STORE

CASB blocked at 14:39:42 — 312MB acknowledged loss

14:38:04 UTC

Impossible travel · 12 identities flagged

Step-up MFA enforced via Okta policy ALT-CTOC-IM-08

14:36:12 UTC

Process injection on WS-441

winword.exe → cmd.exe → rundll32.exe chain

14:35:47 UTC

Phishing wave — 142 messages held

Proofpoint signature match · auto-quarantine engaged

14:32:11 UTC

sqlcmd outside maintenance · PROD-DB-01

Awaiting analyst review

14:24:09 UTC

Wire velocity anomaly · EMEA corridor

\$4.2M aggregate · 11 transactions held by fraud ops

14:21:44 UTC

CVE-2026-28114 added to KEV

Edge appliance — exposure window 6h 14m

14:18:02 UTC

Playbook PB-INC-44 completed

CMP-2836 contained — 8 actions, 0 manual interventions

LIVE ALERT STREAM

CMP-2841 SCOPE

LIVE

14:42:08

CRITICAL

Suspicious LSASS access — credential dump patt...

EDR/Carbon Black · APP-77.cti.prod

14:41:52

CRITICAL

DCSync detected from non-DC origin

NDR/Vectra · DC-01.cti.prod

14:39:18

CRITICAL

Anomalous egress — 2.3GB to unsanctioned dom...

CASB/Netkope · OBJ-STORE.cti

SIGNALS

TOP CONTRIBUTORS

BY DETECTION SOURCE

EDR/Carbon Black

62%

NDR/Vectra

48%

IDP/Okta

37%

CASB/Netkope

29%

SIEM/Splunk corr.

22%

Fraud/Acme

14%

TOP IOCS · LAST HR

IP 185.244.31.0/24

DOMAIN cdn-msft-update[.]com

HASH a1c4..f7e9 (loader.dll)

CERT-SHA1 3c91..0b22 (revoked)

01 KPI strip

Seven headline metrics with deltas against the rolling baseline. The critical-incident tile is reserved for the worst number, color used sparingly so when it appears, it's read.

03 Active campaigns table

Bayesian-fused groupings of alerts that appear to belong to the same actor. Each row links to topology, MITRE mapping, and the live alert feed.

05 MITRE ATT&CK heatmap

02 Cyber Kill Chain strip

The seven canonical attack stages with per-stage event volume, active correlations, and detection confidence. A stage with active correlations gets a red gradient, drawing the eye without alarm-fatiguing the analyst.

04 Blast-radius topology

Assets compromised by the selected campaign, with edges colored by intent: solid red for exfiltration, coral for escalation, dashed for lateral movement. Pulsing nodes are still actively involved.

06 Timeline · live feed · signals

The seven primary tactic columns with the techniques observed in this campaign highlighted. Tactics with critical-confidence hits go deep red. Click any technique to filter the feed.

Three coordinated panels: a human-readable narrative of cross-stream events, a live alert stream filtered to the campaign, and the top contributing detection sources and IOCs.

"The Fusion Kill Chain is the only screen where an analyst can answer 'what's happening?', 'where is it?', and 'what should I do next?' without clicking, and click into any one of them to investigate."

Why a kill chain at all?

The Lockheed Martin Cyber Kill Chain isn't just an industry convention, it's a *shared language*. By laying every alert against the same seven stages, the dashboard makes correlation legible to a tier-1 analyst, a fraud investigator, and a CISO simultaneously. Each persona reads a different layer of the same diagram, but they're looking at the same picture.

KILL CHAIN · ATTACK PROGRESSION

Lockheed Martin's seven-stage model of how a sophisticated cyberattack unfolds: Reconnaissance → Weaponization → Delivery → Exploitation → Installation → Command & Control → Actions on Objectives.

MITRE ATT&CK · TECHNIQUE TAXONOMY

An open matrix of adversary *tactics* (the why) and *techniques* (the how). Mapping detections to ATT&CK lets defenders communicate precisely. Identifiers look like T1059.

FUSION · CROSS-STREAM CORRELATION

Combining alerts from independent detection systems into unified campaigns. A phishing email plus an unusual login plus an egress spike means little alone, fused together, they're an intrusion.

IOC · TTP · CAMPAIGN

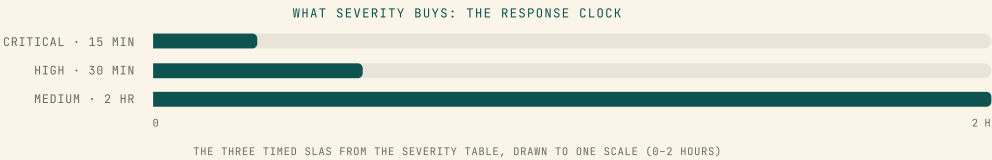
An **IOC** is observable evidence (IP, hash, domain). **TTPs** are adversary behavior, harder to evade. A **campaign** is a correlated set of activity attributed to one actor, the unit of investigation.

07 · THE SEVERITY SYSTEM

Five levels. Color, label, position, motion. *Never just color.*

Color-blind safe severity required encoding the level in more than one channel. Every severity is a *compound signal* a colored pill plus a label plus, in critical cases, a pulsing motion cue. The system never relies on red alone to communicate critical.

SEVERITY	COLOR	DEFINED AS	SURFACE TREATMENT
■ CRITICAL	■ #b91c1c	Active compromise, exfiltration in progress, or known-exploited vulnerability on an exposed asset. SLA: 15 min response.	Pulsing motion, top of queue, paged to T3
■ HIGH	■ #e64d3c	Strong correlation with active campaign, escalation indicator, or significant policy violation. SLA: 30 min response.	No motion, prominent surface treatment
■ MEDIUM	■ #d4a036	Investigative state, likely benign but warrants analyst review. SLA: 2 hr.	Standard row, no special emphasis
■ LOW	■ #1570ef	Informational. Low-confidence detection or background context.	Collapsed by default in dense views
■ RESOLVED	■ #16a34a	Successfully contained, false positive confirmed, or remediated. Visible for trend reporting; never noisy in live queue.	Faded surface, archived after 24h



Type as data

Numbers dominate this product. We selected **Inter** for display headlines (clear at distance for wallboards) and **JetBrains Mono** for all numeric data, with tabular-numeral spacing. Numbers in the same column always align vertically, a tiny detail that compounds across a 1,400-alert queue.

Body type is **DM Sans** at 13–14 px. We rejected the temptation to go smaller. Density comes from *tight row padding* (6 px) and economical labels, not from font size, analysts read this screen for an entire shift.

08 · THE TRIAGE WORKFLOW

The drawer that carries context with the analyst.

When an analyst clicks any alert anywhere in the system, the same right-side drawer slides in. It contains the six-step triage workflow, the alert context, the related signals from the last thirty minutes, recommended playbooks ranked by confidence, and a notes field, all without leaving the dashboard the analyst was on.

CTOC · COMMAND

VIEW / OPS / FUSION KILL CHAIN

RANGE LIVE 1H 24H 7D 30D

SEARCH

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Fusion Kill Chain

ACTIVE CAMPAIGNS · CROSS-STREAM CORRELATION · MITRE ATT&CK ALIGNMENT 3 CRITICAL · 4 HIGH

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7 ▲ +2

2 escalated past 06:00 UTC

OPEN ALERTS

218 ▼ -14

1,442 ingested last hr

CRITICAL INCIDENTS

3 ▲ +1

INC-9981 ETA contain 14:55

ASSETS COMPROMISED

12 ▲ +3

7 production · 3 corp · 2 cloud

MEAN TARGET

LOCKHEED CYBER KILL CHAIN

STAGE 5 SELECTED · INSTALLATION

STAGE 01

Events

142

STAGE 02

Events

73

STAGE 03

Delivery

87

STAGE 04

Exploitation

14

STAGE 05

Installation

2

ACTIVE CAMPAIGNS

CORRELATED · LAST 72H

7 ROWS FILTER

ID

CAMPAIGN

ACTOR

STAGE

SEV

CONF

ASSETS

MTTD

CMP-2841

Operation TIDEWAVE

APT-441 (Nostramo)

6/7

CRITICAL

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81%

3

00:11:55

CMP-2836

Operation PALEHORSE

Unattributed

4/7

HIGH

68%

2

00:24:08

CMP-2832

Operation IRONBLOOM

eCrime cluster

3/7

MEDIUM

55%

12

01:02:44

CMP-2829

Operation HEARTSTONE

APT-209 (Vanta)

2/7

MEDIUM

47%

0

02:14:01

CMP-2825

Operation CLEARWATER

Insider (suspected)

5/7

CRITICAL

74%

1

04:08:20

CMP-2822

Operation SILVERMOTH

FIN-318

3/7

LOW

33%

0

—

MITRE ATT&CK · ENTERPRISE

CMP-2841 MAPPED · 18 TECHNIQUES OBSERVED

ALT-78441 · EDR/Carbon Black

X CLOSE

Suspicious LSASS access — credential dump pattern

CRITICAL STAGE 5/7 CONF 92% OPEN CMP-2841

TRIAGE WORKFLOW

DETECT TRIAGE INVESTIGATE ESCALATE CONTAIN RESOLVE

BACK ADVANCE INVESTIGATE ASSIGN ESCALATE

ALERT CONTEXT

Host

APP-77.cti.prod

First seen

2026-05-26 14:42:08 UTC

Detector

EDR/Carbon Black

Stage

5/7 · Installation

Confidence

92% (fusion model v3)

Assigned

M. Okafor

Campaign

CMP-2841

SLA

15 min (P1)

RELATED SIGNALS (LAST 30M)

TIME

SOURCE

SIGNAL

SCORE

14:42:08

EDR

LSASS handle acquired (T1003)

92

14:41:52

NDR

DCSync RPC pattern

89

14:39:18

CASB

Egress 2.3GB to cdn-msft-update[.]com

94

14:36:12

EDR

winword.exe → cmd.exe injection

78

14:24:11

IDP

MFA bypass attempt (svc-acct)

67

RECOMMENDED PLAYBOOKS

PB-CRED-12

Credential access — isolate + force rotation

91%

RUN 3.2M

PB-EXFIL-04

Exfiltration — egress block + DLP retro-scan

84%

RUN 4.8M

PB-IDP-22

Identity — invalidate sessions, enforce step-up

79%

RUN 1.4M

ANALYST NOTES

Add observation, IOC, escalation reason...

VIEW IN SIEM

DOWNLOAD EVIDENCE

CREATE CASE

MARK FALSE POSITIVE

SAVE & CLOSE

THE SIX STEPS

- 01 · DETECT

An alert fires from a detection source. The clock starts on MTTD.
- 02 · TRIAGE

Analyst confirms severity, dedupes, decides whether to escalate or close.
- 03 · INVESTIGATE

Pull related signals, build the narrative, identify scope.
- 04 · ESCALATE

Hand off to a higher tier or adjacent function (fraud, legal, HR).

SOAR · ORCHESTRATION, AUTOMATION & RESPONSE

Tooling that runs *playbooks* pre-defined response sequences. The drawer recommends playbooks ranked by Bayesian confidence and lets the analyst kick them off without leaving the alert.

PLAYBOOK

A pre-built response sequence: e.g. *PB-CRED-12* (isolate host, revoke sessions, force credential rotation, retro-scan adjacent assets). Each reports its success rate, mean execution time, and confidence over time.

05 · CONTAIN	Execute response playbooks. Isolate hosts, revoke sessions, block egress.
06 · RESOLVE	Close the case, capture lessons learned, push detection improvements upstream.

SLA · SERVICE-LEVEL AGREEMENT

The internal commitment to respond within a defined window for each severity, 15 minutes for Critical, 30 for High, 2 hours for Medium. SLA timers are surfaced on every incident.

SIEM · SECURITY INFO & EVENT MGMT

The platform that aggregates and indexes log data from every system. The dashboard sits on top of the SIEM, the SIEM is the system of record, the dashboard is the system of attention.

09 · THE DESIGN SYSTEM

One kit of parts *behind all thirteen screens.*

Thirteen dashboards across three tiers could only feel like one product if they were *assembled*, not authored. So the system is a token layer, color, type, spacing, elevation, motion, plus a library of composable components defined once and reused everywhere. A severity pill, a KPI tile, a data table, the triage drawer: each means the same thing on every screen.

What changes between screens is **data and density**, never the grammar. That is how new domains, Insider Threat, Money Movement, Executive Travel, shipped without re-deciding the basics, and how the product stayed coherent as it tripled in scope.

FOUNDATIONS

Type as data

Inter for wallboard-legible headlines, **DM Sans** for 13–14px body, **JetBrains Mono** for every number and ID, with tabular numerals so columns align down a 1,400-row queue.

1,432 ▾ 12% OPEN ALERTS · 24H

Restraint & density

A monochrome base on near-black (#0b1016). Density comes from 6px row padding and economical labels, not shrinking type. Color is rationed, when it appears, it is a signal, not decoration, so an analyst reading for a full shift never tunes it out.

Severity scale

Five compound signals, color + label + square, never hue alone (full table in §07). Re-tinted for the dark theme so they read at wallboard distance.

■ CRITICAL

■ HIGH

■ MEDIUM

■ LOW

■ RESOLVED

THE COMPONENT LIBRARY

12 PRIMITIVES

01

Panel

The titled container every widget sits in, a head with title and tools, a body for content.

02

KPI strip

Six-up tabular metrics with deltas and a trailing baseline footnote, atop every screen.

03

Severity pill

The five-level compound signal, identical on every screen, never red alone.

04

Data table

Sticky mono headers, hover rows, right-aligned tabular numbers, graceful truncation.

05

Kill-chain strip

The seven attack stages with per-stage volume, active correlations, and confidence.

06

MITRE grid

Tactic columns with the observed techniques highlighted, click any to filter the feed.

07

Asset topology

Node-and-edge blast radius, edges colored by intent, live nodes pulse.

08

Triage drawer

The right-side flyout with the six-step workflow, invoked from any alert, anywhere.

09

Timeline

A human-readable narrative of fused cross-stream events in chronological order.

10

Live feed

A streaming alert list that highlights new rows as they arrive, then settles.

11

Heatmap

A density grid for coverage, utilization, and time-of-day load.

12

Buttons & chips

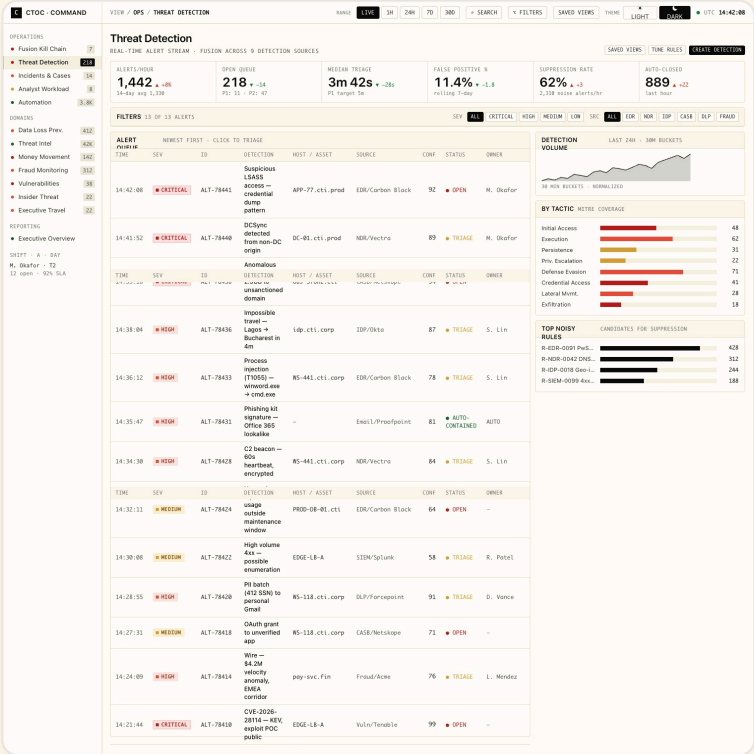
Primary, danger, and ghost actions, plus tags and filter pills, one set, sized for density.

"Twelve primitives and one token file are the reason a thirteen-dashboard product reads as a single screen, designed once, reused everywhere, impossible to drift."

10 · THE OTHER TWELVE SCREENS

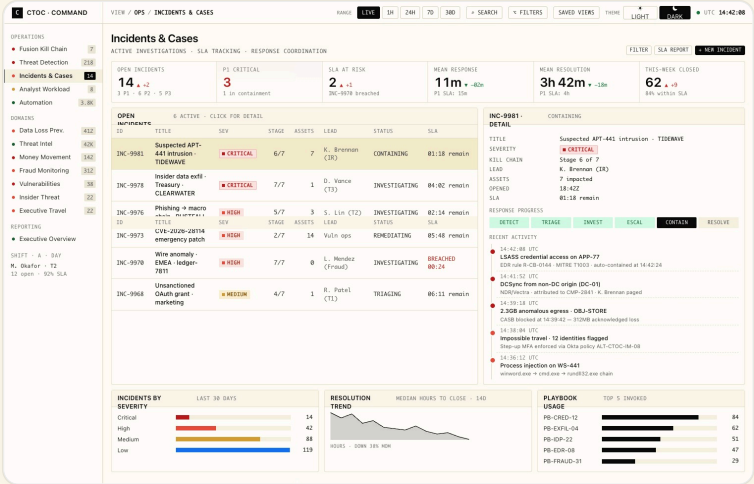
One pattern language, applied across *every operational domain*.

Every dashboard inherits the same chrome (top bar, sidebar, time-range, KPI strip) and the same severity system. What varies is the *information architecture* each domain has a unique payload, but the analyst reads them all the same way.



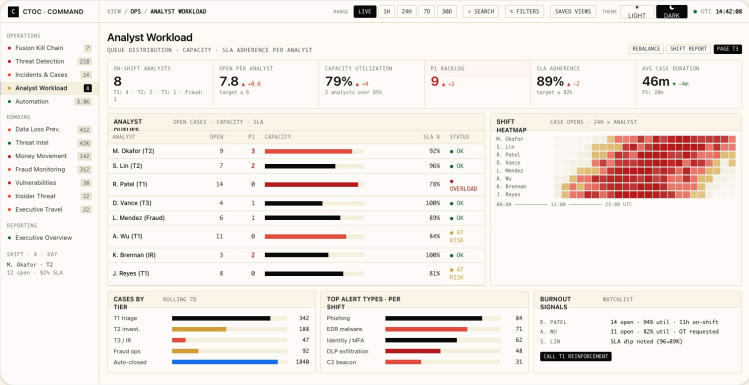
02 Threat Detection

The dense real-time queue. Filters by severity and source; tactic-coverage bars surface which MITRE columns are lighting up; "top noisy rules" gives the analyst direct input into the suppression model.



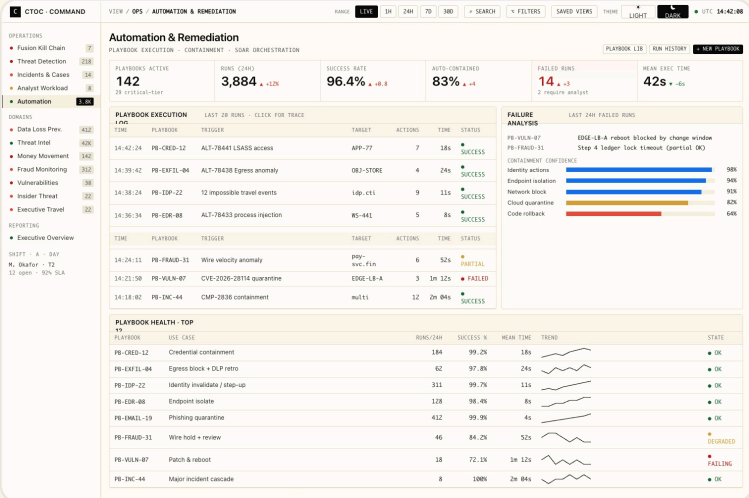
03 Incidents & Cases

Where alerts become investigations. SLA timers color when at risk, breached cases surface to the top, response-progress steps mirror the drawer workflow for continuity.



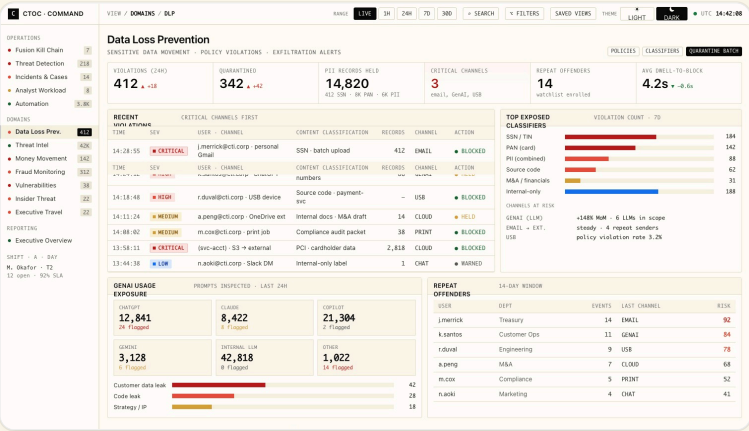
84 Analyst Workload

Manager view. A 24xN heatmap shows where the floor is hot, capacity bars flip color past 90% utilization, and a "burnout signals" panel asks the manager to act on early warning.



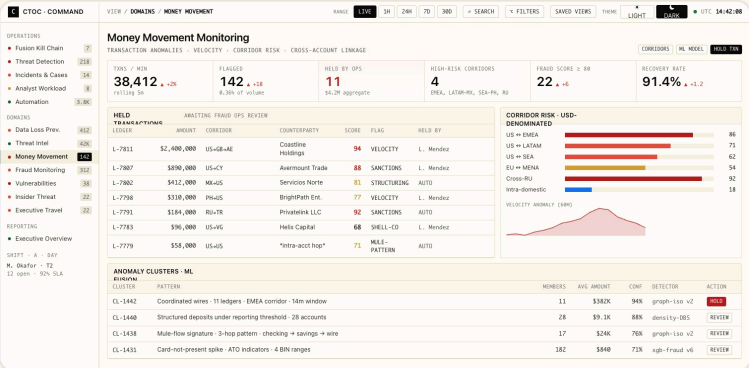
85 Automation & Remediation

SOAR health. Per-playbook sparklines reveal degradation before it becomes a failure; containment confidence is broken out by action type so analysts know what they can trust to auto-run.



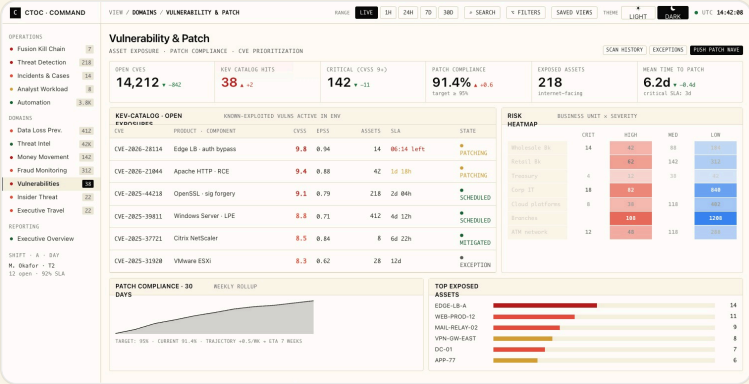
86 Data Loss Prevention

Sensitive-data egress monitoring. We added GenAI prompt inspection as a first-class channel, the fastest-growing exfiltration surface in the bank.



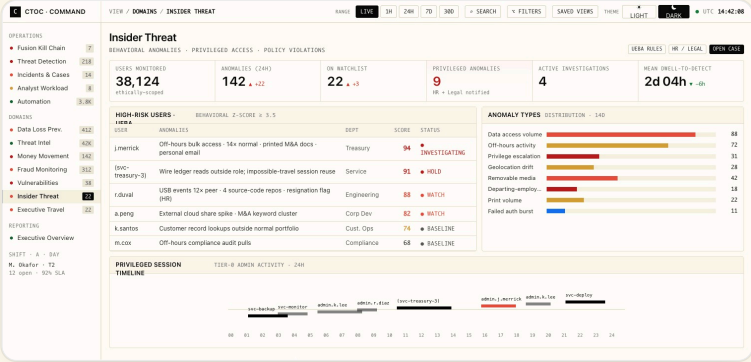
88 Money Movement

Where cyber meets fraud. Transaction-velocity anomalies, corridor risk, ML clustering of coordinated wires, the dashboard the fraud team and the SOC share.



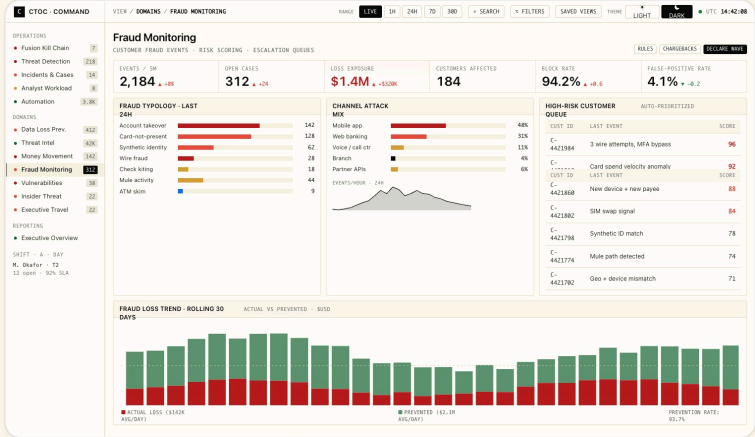
89 Vulnerability & Patch

CVE exposure prioritized by CISA's KEV catalog and EPSS scoring. The business-unit heatmap turns "we have 14,000 CVEs" into "Wholesale Banking has 14 critical."



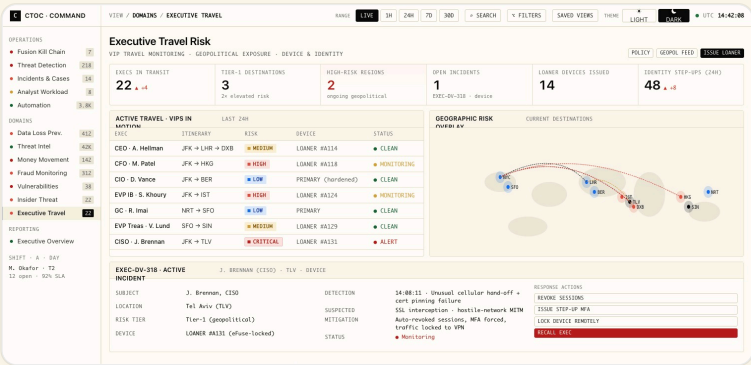
10 Insider Threat

UEBA-driven behavioral anomaly detection with built-in HR and Legal escalation paths. Designed with ethics review baked in, explicit watchlist enrollment, evidence trail, no shadow monitoring.



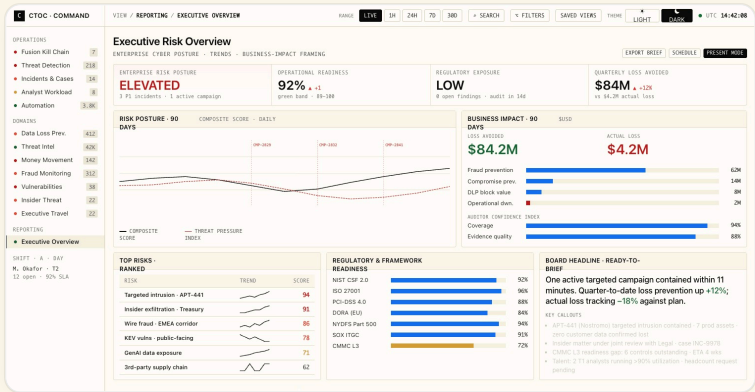
11 Fraud Monitoring

Customer-facing fraud, ATO, card-not-present, mule activity. The loss-trend chart reports prevented vs actual side-by-side: leadership wants to know what we saved, not just what we lost.



12 Executive Travel

VIP travel risk, geopolitical overlays, device tracking, MITM detection. Built after a real incident where a CISO's loaner device hit a hostile network in transit.



13 Executive Risk Overview

The board-ready view. Composite risk trend, dollar-value framing, regulatory-framework readiness bars, and a one-paragraph "headline" the CISO can read into the room verbatim.

11 · GLOSSARY

The shared vocabulary *of the system.*

Cybersecurity is acronym-dense and the product reflects that. We did *not* try to simplify the terminology, analysts speak this way, and dumbing it down would have made the product harder, not easier, to use. Every term has a consistent definition surfaced as a tooltip on first use.

APT
Advanced Persistent Threat

A sophisticated, typically state-sponsored adversary group, long-lived, well-resourced, focused on espionage or financial gain. Identified internally with codenames (e.g. APT-441 "Nostromo").

C2
Command and Control

Stage 6 of the kill chain. The infrastructure an attacker uses to issue instructions to malware once it's installed, typically a periodic encrypted "beacon" call-out.

CVE · CVSS · EPSS
Vulnerability scoring

CVE identifies a vulnerability. **CVSS** scores how bad it could be (0–10). **EPSS** estimates exploit probability in the next 30 days.

ATO
Account Takeover

Fraud where an attacker gains control of a legitimate customer's account through credential theft, SIM swap, or social engineering.

CASB
Cloud Access Security Broker

The control layer between users and SaaS apps. CASB telemetry is a primary detection source for cloud data exfiltration and OAuth abuse.

DCSync
Active Directory attack

An attack where an adversary impersonates a domain controller to harvest credentials from Active Directory. Surfaced as the stage-6 critical alert on DC-O1.

DLP

Data Loss Prevention

Controls that prevent sensitive data from leaving the enterprise, through email, USB, cloud sync, printer, GenAI prompts, etc.

KEV

Known Exploited Vulnerabilities

CISA's catalog of vulnerabilities being actively exploited in the wild. KEV entries get priority patching, typically 14-day SLA.

MFA bypass

Identity attack

Defeating multi-factor authentication, through push-fatigue, SIM swap, OAuth abuse, or token theft. Identity is the new perimeter and MFA bypass the new front door.

P1 / P2 / P3

Incident priority

P1 is critical (15-min SLA). P2 is high (30-min). P3 is medium (2-hour). Priority drives queueing and escalation throughout the system.

SOAR

Security Orchestration & Response

The automation layer that runs response playbooks across tools without human intervention. The Automation dashboard reports SOAR health.

EDR

Endpoint Detection & Response

The agent on every endpoint that monitors process and file behavior in real time. Highest-fidelity detection source in most enterprises.

LSASS

Local Security Authority Subsystem

The Windows process that holds authentication secrets in memory. Accessing it (T1003.001) is a near-certain sign of credential harvesting.

NDR

Network Detection & Response

The detection layer that watches network traffic for anomalies, beacons, lateral movement, exfiltration. Complementary to EDR.

PII · PAN · PCI

Sensitive data classes

PII personally-identifiable information. PAN payment card account number. PCI the cardholder data regulatory regime.

UEBA

User & Entity Behavior Analytics

Machine-learning models that baseline normal behavior per user and entity, then flag statistical deviation. Powers the Insider Threat dashboard.

ANOTHER CASE STUDY

Goals-Driven Finance →

rbabiarz@gmail.com

← Back to portfolio

COLOPHON · BUILD RECORD

VIEW SOURCE ON GITHUB ↗

SITE EST	DEPLOYMENTS	LAST DEPLOY	VERSION	UPDATED				
Jun 22, 2026	108	Jul 18, 2026	v1.11.113	Jul 19, 2026				
BUILT WITH	Anthropic Claude Code · Playwright · Figma + MUI kit · Cursor · Python 3 · React 18 (CDN, no build step) · Git → GitHub Pages							
ARCHITECTURE	29 self-contained pages, ~76,000 hand-authored lines, zero npm dependencies, no bundler — every page is a single file that opens anywhere and deploys as pure static hosting. The interactive layer is bespoke vanilla JS: a photometric engine scored against real parking uniformity targets, an AI AutoLayout demo with ranked candidates, playable decision drills, and client-side search over a Python-generated index. WCAG 2.2 AA target with a standing audit; three-tier design tokens (primitive → semantic → component).							
AI-PAIRED	Human-led, AI-accelerated: designed and engineered in pair with Anthropic Claude Code, with Playwright browser automation verifying every change — the same receipts-over-vibes workflow the case studies argue for. 11 case studies, several of them about designing AI systems people can trust.							
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